NewsBot Intelligence System 2.0 — Technical Documentation

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1. System Architecture Overview

NewsBot 2.0 extends the mid-term system (Modules 1-8) with four advanced modules. The system processes 1,490 BBC News articles through a 12-module pipeline: preprocessing, TF-IDF extraction, POS tagging, syntax parsing, sentiment analysis, multi-class classification, NER, topic modeling, summarization, semantic search, multilingual detection, and a conversational query interface.

Module	Technology	Output
M2: Preprocessing	NLTK tokenize/lemmatize	Clean token lists
M3: TF-IDF	sklearn TfidfVectorizer	5000-feature matrix
M4: POS Tagging	NLTK pos_tag	POS distribution %
M5: Syntax	spaCy dep parser	SVO triples
M6: Sentiment	VADER + TextBlob	Compound scores
M7: Classification	Linear SVM pipeline	92%+ accuracy
M8: NER	spaCy en_core_web_sm	PERSON/ORG/GPE entities
M9: Topic Model	sklearn LDA (5 topics)	Topic-article mapping
M10: Summarization	TF-IDF sentence scoring	30% compressed summaries
M11: Multilingual	langdetect + deep-translator	Language labels + EN translation
M12: Conversational	Rule-based NLU + cosine search	Natural language answers

2. Module 9: Advanced Topic Modeling (LDA)

Latent Dirichlet Allocation (LDA) is an unsupervised machine learning technique that discovers hidden thematic structure in a collection of documents. Unlike supervised classification which requires pre-labeled categories, LDA automatically finds co-occurring word patterns that represent underlying topics. I trained LDA with n_components=5 using the online learning method for computational efficiency on Google Colab free tier. The model processes a CountVectorizer matrix of 2,000 features with min_df=2 and max_df=0.85 to exclude both rare and overly common words.

The topic-to-category heatmap reveals that LDA topics closely align with BBC categories, validating that the unsupervised approach discovers the same structure as human-annotated labels. This cross-validation between supervised and unsupervised methods strengthens confidence in both the data quality and the model's learning.

3. Module 10: Text Summarization & Semantic Search

I implemented extractive summarization using TF-IDF sentence scoring. Each sentence in an article is vectorized using TF-IDF, and sentences are ranked by their aggregate importance score. The top N sentences (default=2) are selected and returned in their original order to maintain narrative coherence. This approach achieves approximately 30% compression ratio — summaries average 30 tokens versus 100+ token originals.

Semantic search uses cosine similarity between a query TF-IDF vector and all article vectors. Unlike keyword search which requires exact word matches, semantic search returns articles that are conceptually similar to the query, even if specific words differ. The search index is pre-built on all articles for fast retrieval.

4. Module 11: Multilingual Intelligence

Language detection uses the langdetect library, which applies statistical models trained on 55 languages to identify the language of any text input. The BBC News dataset is predominantly English (99%+), as expected for a British broadcaster. The translation pipeline uses deep-translator (Google Translate API wrapper) to convert non-English articles to English before classification, enabling global news monitoring.

5. Module 12: Conversational Interface

The conversational interface uses rule-based Natural Language Understanding (NLU) combined with cosine similarity search. User questions are parsed for intent signals: category keywords map to BBC categories, sentiment words filter by positive/negative tone, count keywords trigger aggregate queries, and summary keywords activate the summarization module. When no rule matches, the system falls back to semantic search for open-ended queries.

6. Performance Results

Metric	Value	Notes
Dataset size	1,490 articles	BBC News, 5 categories
Classification accuracy	92%+	Linear SVM, test split
Avg summary compression	~30%	Extractive, 2 sentences
Languages detected	2-3 languages	Predominantly English
LDA topics	5 topics	Matches 5 BBC categories
Search response time	<1 second	Pre-built TF-IDF index